What You Are Actually Measuring When You Benchmark Speculative Decoding Under FP8

A measurement case study on vLLM 0.29.0 and NVIDIA L4

Akshath Tiwari
Independent
tiwariakshath@gmail.com

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Abstract

Speculative decoding and FP8 quantization are standard in LLM serving, and their interaction is usually reported from single runs. Measuring it on vLLM 0.29.0 and an NVIDIA L4, we found four confounds larger than the effect. First, attention-backend selection is conditioned on KV-cache dtype, so an FP8-versus-BF16 comparison under defaults also swaps the attention kernel, and target and draft select independently. Second, speculative throughput falls into two disjoint regimes: across ten L4 sessions on Modal, with the ratio of CUDA-graph to `--enforce-eager` throughput measured in the same container, three sessions sit at 0.487–0.768 and seven at 1.003–1.039, with nothing between. An intervention on a rented L4 places the cause in the gVisor sandbox that Modal runs GPU functions in: on one card, vLLM 0.29.0 gives 0.736 under gVisor and 1.018 under Docker's default runtime, vLLM 0.30.0 is at parity in the same sandbox (1.040), and in the deficit the GPU is idle about a third of the time, waiting on the host. Modal attributes the parity of its later sessions to a gVisor change that backs GPU-shared memory with huge pages; under the public release containing it the deficit persists on a second card, but the host never allocated such pages, so that explanation remains untested. Third, acceptance length τ has a boot-to-boot coefficient of variation of 0.62%, and below 0.88% in every session, which we could not find reported. Fourth, inter-token latency computed from stream arrivals is inter-chunk latency: a speculative step emits every accepted token in one chunk, so a p95-ITL objective scores the highest-throughput configuration in our grid at zero goodput. The first three add error; the fourth reverses the ranking. We release the harness, the raw records and every correction we made.

1 Introduction

Speculative decoding trades a small draft model's cheap guesses against the target model's expensive verification [1, 2]. Its benefit is governed almost entirely by one quantity: τ, the mean number of tokens emitted per verification step (accepted draft tokens plus the bonus token). We call this *acceptance length* rather than an acceptance *rate*, since $\tau > 1$ by construction and is not a proportion. FP8 quantization independently reduces weight and KV-cache footprint, which matters because KV capacity is what binds under concurrency [3]. Serving stacks ship both. The obvious question — does quantization degrade acceptance, and what does the combination buy under load — is asked frequently and answered from single-run measurements.

We could not answer it. Four separate times, a result we were about to report turned out to be produced by something other than the variable under study. This paper reports those four things, because they are larger than the effect they obscured.

Contributions.

1. **Backend substitution under dtype change** (§3). vLLM's `auto` backend selection is conditioned on KV-cache dtype, and target and draft select independently. A precision A/B under defaults is a kernel A/B as well.
2. **A sandbox-dependent throughput regime** (§4). Identical configurations disperse across boots at constant acceptance and fall into two disjoint regimes: ratio 0.487–0.768 with graph-arm CV 3.73–13.92%, or 1.003–1.039 with CV 0.82–1.95%, with nothing between. On one rented card the deficit appears under gVisor and not under runc, follows the engine release and not the driver, and comes with a GPU idle a third of the time, on two cards. Power throttling, CPU starvation, the driver and one candidate engine change are each ruled out by measurement. The platform names a hugepage change in its sandbox as what restored parity; we could not exercise it off the platform. `--enforce-eager` removes the variance but is not a free control — it costs FP8-weight configurations a fifth to a half of their throughput and cannot boot speculative FP8-KV cells at all on this card.
3. **Acceptance dispersion** (§5). τ has CV 0.62% across 12 fixed-prompt boots. We withdraw two claims of our own that rested on one boot per arm, and decline to convert the dispersion into an equivalence bound for a comparison measured at a different operating point.
4. **ITL is inter-chunk latency** (§6). A speculative step emits every accepted token in one stream chunk, so a p95-ITL objective ranks the highest-throughput configuration in the grid as fully non-compliant. This one does not add error — it inverts the decision.
5. **Measurements after diagnosing four confounds** (§7): compatibility, KV capacity, task accuracy at n=256, and goodput. Three of the four are controlled; draft-side backend selection is not, because the engine offers no way to pin it.
6. **A fully auditable artifact** (§10): append-only raw records, script-regenerated derived tables, provenance checked by a script that fails the build, and every retraction preserved rather than removed.

We emphasise the last because it is what made the others findable. Most of our corrections were caught by re-reading stored records against a hypothesis formed later — impossible if results are summarised at write time.

2 Setup

L4 was chosen deliberately: it is the compute capability named in the upstream reports that motivated the study, and at 22 GiB the KV-capacity constraint actually binds, which it does not on an 80 GB card.

Engine	vLLM 0.29.0, pinned by image; §4 adds a 0.30.0 replication
GPU	NVIDIA L4 (SM89, Ada, 24 GB nominal / 22.03 GiB usable), rented per second
Target	Qwen3-4B
Draft	`z-lab/Qwen3-4B-DFlash-b16`, block size 16, 5 speculative tokens
Also	n-gram (prompt-lookup) speculation, no draft checkpoint
Workload	GSM8K [4], sampled without replacement
Base image	`nvidia/cuda:13.0.3-devel-ubuntu24.04`

The harness never imports the engine. It launches vLLM as a separate server process and communicates with it over HTTP. This prevents the experiment from using an in-process path unavailable to deployments, and it includes server-side request handling, scheduling, batching and streaming. It does *not* reproduce every production environment, and it does not eliminate client or network overhead.

Every measurement is content-addressed. A configuration hashes to a `cell_id`; identical configuration yields an identical id on any machine. Records are append-only, so a re-run supersedes rather than overwrites and the superseded record remains readable. Several of our corrections depend on that history being intact.

Definition. We define

$$\tau \;=\; 1 + \frac{\text{accepted draft tokens}}{\text{verification steps}}.$$

The +1 is the bonus token, which vLLM's counter excludes. Omitting it understates τ by exactly 1 and silently changes every speedup ratio derived from it.

3 Confound 1: the attention backend moves when the dtype moves

vLLM's `auto` attention-backend selection is a function of KV-cache dtype. At BF16 KV it selects FlashAttention-2; at `fp8_e4m3` it selects FlashInfer. An experimenter varying only precision therefore varies the attention kernel too, and any resulting difference is jointly attributable.

It is worse than one substitution. Target and draft models select backends *independently*, at two distinct call sites, so pinning the backend affects the target while the draft continues to select on its own. A pinned-backend precision comparison is still confounded on the draft side for speculative cells, which we record per cell rather than assert away.

Compatibility is not uniform. Of the four attention backends vLLM 0.29.0 offers on SM89, FP8 KV runs on exactly two.

Table 1: FP8 KV cache support by attention backend, Qwen3-4B + DFlash, SM89. `launch_failed` is rejection at configuration time.

Backend	FP8 KV status	KV tokens
FLASHINFER	ok	115,920
TRITON_ATTN	ok	129,584
FLASH_ATTN	launch_failed	—
FLEX_ATTENTION	launch_failed	—
`auto`	ok (selects FLASHINFER)	115,920

This is the mechanism behind the dtype-conditioned selection: `auto` does not switch away from FlashAttention-2 under FP8 KV as a preference, it switches because FlashAttention-2 *cannot* serve FP8 KV on this hardware. The original hypothesis motivating this study — that FlashInfer's FP8 attention kernels are SM90-only, so FP8 KV plus non-causal drafting would crash on Ada — is refuted. The combination runs.

Our own error here. We initially concluded that the backend selection flag was accepted, echoed back, validated against, and then ignored, and reported this upstream. That was wrong: the flag controls the target, and what we observed was the draft selecting independently. We corrected the report. The mistake is instructive because we compared requested-versus-selected backend for *a server* when there is no such thing as a server's backend — there are two.

4 Confound 2: speculative throughput falls into one of two regimes, and the sandbox decides which

Booting an identical speculative configuration repeatedly produces widely scattered throughput while acceptance stays put — in some container sessions. In others the same configuration is quiet and reaches parity with its own eager control. We ran eleven sessions of the identical sweep on Modal, a serverless GPU platform, ten on the L4 and one on an A10, each landing on a distinct physical card:

		default (graph)		eager control			
Engine	Session	tok/s	CV	tok/s	CV	ratio	Regime
0.29.0	09-23	39.2	**13.92%**	80.5	1.44%	**0.487**	deficit
0.29.0	09-29 a	61.2	**3.73%**	79.7	1.69%	**0.768**	deficit
0.29.0	10-04	56.6	**10.38%**	78.1	6.25%	**0.724**	deficit
0.29.0	10-31 a	81.2	1.53%	80.9	1.24%	1.003	parity
0.29.0	10-31 b	82.0	1.00%	81.1	1.14%	1.011	parity
0.29.0	10-31 c	81.2	1.95%	80.0	1.02%	1.015	parity
0.29.0	11-01	83.5	0.82%	82.5	0.75%	1.012	parity
0.30.0	09-29 b	82.8	0.99%	81.6	1.28%	1.014	parity
0.30.0	09-29 c	78.2	1.61%	77.0	2.79%	1.017	parity
0.32.0	10-31 d	81.7	0.86%	78.6	3.12%	1.039	parity
0.29.0	09-30 (**A10**)	132.6	2.08%	138.3	1.28%	0.959	parity

Each cell is 12 fixed-prompt boots inside a single container, except three truncated by a runner timeout (09-23 eager $n = 8$; 09-29 c and 10-04 eager $n = 11$). Every session ran on a distinct physical card, recorded by UUID, except 09-23, which predates that recording. Every L4 card reports 22.5 GiB, and every one whose limits were recorded (all but 09-23) reports the same 2040 MHz clock ceiling and 72 W power limit.

The split is binary, and it is one phenomenon rather than two. The ratio column has no intermediate value: three sessions between 0.487 and 0.768, seven between 1.003 and 1.039, and a gap of 0.235 between the regimes wider than the spread within either. The graph arm's dispersion partitions the same way — 3.73–13.92% in the deficit sessions against 0.82–1.95% in the parity sessions, again disjoint. **The two partitions are identical.** Under random

assignment, the probability that the three lowest-ratio sessions are exactly the three highest-dispersion sessions is $1/\binom{10}{3} = 0.008$. We therefore stop reporting the deficit and the dispersion as separate results: a session is in one regime or the other, and in the deficit regime it is both slow and unstable.

The separation holds boot by boot, not only session by session. Pooling every fixed-prompt default-arm boot at concurrency 1 across all ten L4 sessions gives $n = 128$: 44 boots from the four sessions carrying the deficit signature, spanning 29.5–65.6 tok/s, and 84 from the seven parity sessions, spanning 75.5–83.9. The fourth deficit session is a 10-04 run that measured no eager arm, so it has no ratio and does not appear in the table above; we include its boots here rather than discard a measured boot for lacking a control it never had. **Not one boot falls between 65.6 and 75.5 — the two groups do not overlap at all**, and the 9.9 tok/s gap between them is 1.7 times the largest gap inside either group (5.7). Every individual boot measured in a deficit session is slower than every individual boot measured in a parity session. A reader who runs this configuration once gets a number from one of two separated populations, and which one depends on things that reader did not choose — and, on a serverless platform, cannot see.

What the split depends on. Earlier readings of this split, first as a regression in vLLM 0.29.0 and then as a random session-scoped regime, compared sessions across two host drivers without knowing it. **Every session up to 4 October ran NVIDIA driver 580.95.05; every session from 31 October ran 610.57.04.** Every deficit session on Modal shares one configuration, and every Modal session that differs from it in exactly one factor is at parity:

Factor changed	To	Sessions	Deficient
none: Modal, L4, 580.95.05, 0.29.0, DFlash		3	**3**
engine release	0.30.0	2	0
host driver	610.57.04	4	0
card	A10	1	0
speculation mechanism	n-gram*	2	0
platform	**GCP, under gVisor**	**1**	**1**
platform and runtime†	GCP, runc	3	0

*Not a one-factor change; see the n-gram paragraph below. †Two factors: Modal runs GPU functions under gVisor, and these ran under Docker's default runtime. Both GCP rows use driver 580.95.05 installed by hand. Within one driver, the engine release and the card each separate the regimes: on 580.95.05, vLLM 0.29.0 is deficient in 3 sessions of 3 against 0 of 2 for 0.30.0, and the L4 in 3 of 3 against 0 of 1 for the A10. n-gram does not, because it changes more than the mechanism. The engine logs see none of it: across 140 graph-arm logs, not one message separates Modal's deficit configuration from any of its neighbours.

The driver alone does not reproduce it. Driver and calendar date are collinear in the Modal records, so no analysis of them can separate the two; only an intervention can. We rented a GCP L4, installed 580.95.05 from NVIDIA's own installer, and ran the unmodified harness inside a rebuild of the Modal image under Docker's default runtime, runc, matching torch (2.13.0+cu130), vLLM, the CUDA base image, the model runner and async scheduling.

The ratio is **1.001** (conservative 95% interval [0.989, 1.013]) with a graph-arm CV of **0.91%**: parity on both signatures, under the configuration that was deficient on Modal three times in three. A decision rule committed before the eager arm was measured said that parity ends the run there, and it did.

Nor does one starved CPU. Every GCP engine log reports torch starting at two threads; no Modal log does, so Modal's containers saw one CPU or less. Pinned to one vCPU, the GCP container reproduces that observable — torch at one thread, the log line absent — and the ratio is **1.028** ([1.012, 1.044]), graph-arm CV **0.95%**. If anything the graph arm gains, which is what CUDA graphs are for: they remove CPU launch overhead, and that overhead costs more when CPU is scarce. A fractional CPU quota, which throttles harder than a pinned core and which torch's thread count cannot tell apart from one, was not tested. Nor is a starved CPU needed: under gVisor the deficit appears with all four vCPUs visible.

A power reading is the obvious candidate: the L4 is a **72 W** part running at 2040 MHz, and a card on that budget could throttle under the graph path and not under eager. **We measured it, and it is refuted.** Sampling achieved clocks, power draw and the driver's own throttle reasons across the measurement window gives, in the 10-04 deficit session, a default arm power-capped for 84% of samples at 1991 MHz and 67.2 W producing 56.6 tok/s, against an eager arm capped for **95%** of samples at **1645 MHz** and **69.8 W** producing **78.1** tok/s. Thermal throttling is 0.0% throughout. The faster arm is capped more, draws more, and clocks lower — the opposite of the prediction.

The same signature now separates the regimes, not just the arms. Instrumenting the parity sessions lets us compare the deficit graph arm against a healthy graph arm running the identical configuration:

Regime	Session	tok/s	SM clock	capped	power	util
deficit	10-04	56.6	**1991 MHz**	0.84	67.2 W	—
deficit†	10-04 b	62.4	**1985 MHz**	0.91	68.7 W	—
parity	10-31 a	81.2	1700 MHz	0.96	70.1 W	91.3%
parity	10-31 b	82.0	1808 MHz	0.96	69.9 W	91.4%
parity	10-31 c	81.2	1777 MHz	0.95	70.0 W	90.5%
parity	10-31 d	81.7	1714 MHz	0.96	70.1 W	90.5%

Default arm only; means over 12 boots. †10-04 b ran no eager arm, so it has no ratio and is excluded from the regime tables; its level and clock put it in the deficit regime unambiguously.

The deficit arms hold a clock 177–291 MHz higher than any parity arm while producing 23–31% fewer tokens, drawing less power, and being capped less often. A card that boosts higher, draws less and delivers less work is not against a ceiling — it is doing nothing for part of the time. Power throttling is refuted as the mechanism rather than merely unsupported, and it is refuted twice over: across arms within a deficit session, and across regimes on the same arm.

What this still does not settle. The stall reading predicts low GPU utilisation in a deficit session. We added `utilization.gpu` sampling on 31 October, and no Modal deficit session has occurred since — all four instrumented Modal sessions landed in the parity regime, where

utilisation is 90.5–91.6% and the two arms are indistinguishable. The sandboxed deficit below supplies the measurement Modal no longer could.

The ratio is the quantity to report. Session 09-29 c ran 5.6% slower than 09-29 b on *both* arms (default 82.8 to 78.2, eager 81.6 to 77.0). Because both moved together, the eager arm identifies that as a property of the card rather than of the execution path, and the ratio absorbs it. Without a control measured in the same container it would have read as a 5.6% replication failure.

Under the sandbox, the deficit reproduces. Modal confirmed (support correspondence, 3 November 2026) that its GPU functions, not only its sandboxes, run under gVisor, whose GPU access goes through `nvproxy`, a proxy that forwards every driver call and is built against each driver version's ABI;[1] that the 4 October containers ran its own gVisor build on 580.95.05 with `--nvproxy-allow-unsupported-driver`; and that library thread counts default to one. Upstream gVisor release-20260914.0, published on 16 September, registers 580.95.05 in exactly that way, as an unsupported ABI derived from 580.65.06. We installed it on a rented L4 and ran the same image on one card, one container per row, changing one factor at a time. The rule for the first row was committed before the machine existed; the other five are exploratory, each with its reading committed before it ran.

Runtime	Driver	Engine	Graph	Eager	Ratio [95%]	Graph util
gVisor 09-14	580.95.05‡	0.29.0	57.2	77.7	**0.736** [0.722, 0.750]	62.2%
runc	580.95.05	0.29.0	82.6	81.1	1.018 [1.004, 1.032]	92.3%
gVisor 09-14	580.95.05‡	0.30.0	81.4	78.3	1.040 [1.022, 1.058]	91.7%
gVisor 09-14	580.95.05‡	0.29.0+pin§	56.7	78.1	**0.726** [0.715, 0.738]	62.7%
gVisor 09-14	610.57.04	0.29.0	58.5	77.5	**0.755** [0.742, 0.767]	63.2%
gVisor 10-26	610.57.04	0.29.0	57.8	77.6	**0.744** [0.732, 0.757]	62.7%

Throughput in tok/s, 12 boots per arm, threads at one, DFlash at concurrency 1; the interval is the conservative one used above. ‡Unsupported ABI: nvproxy logs that the version "is not officially supported". §vLLM 0.30.0's pin of FlashInfer's host arange buffer, backported into 0.29.0 and nothing else.

Under runc the card is at parity. Under gVisor, vLLM 0.29.0 loses 30% of its graph-arm throughput and 4% of its eager-arm throughput, and the ratio lands inside Modal's deficit band. vLLM 0.30.0 in the same sandbox is at parity, as it was on Modal on the same driver. The driver does not matter here: on 610.57.04, which nvproxy supports, and under a late-October gVisor, 0.29.0 is still deficient. **The telemetry completes the stall reading.** The deficit graph arm is busy 62.2–63.2% of the time against 90.0–92.3% at parity, at 2020–2024 MHz against 1610–1893 MHz and 67.5–68.0 W against 69.8–70.2 W, the same signature as Modal's 10-04 session: for about a third of every step the GPU is waiting on the host. A per-call proxy cost would hurt the eager arm, which launches every kernel from the host, more than the graph arm. The opposite happens, so the cost lies in something the 0.29.0 graph path does and its eager path does not. One candidate, a pageable host-to-device copy that vLLM 0.30.0 pins, changes nothing when its pin is backported (0.726).

What the sandbox does not explain. Two things. Modal's deficit sessions scattered from boot to boot (graph-arm CV 3.73–13.92%); the sandboxed deficit here does not (0.91–1.73%),

[1] https://gvisor.dev/docs/user_guide/gpu

so on Modal something else, plausibly a shared host, added the dispersion. And four Modal sessions on 610.57.04 with vLLM 0.29.0 are at parity (1.003–1.015) while sharing every factor we can record with the deficit sessions, the sandbox included.

Modal's explanation for the second (support correspondence, 4 November 2026) is that its late-October gVisor build lets memory shared with the GPU use huge pages, through two changes now upstream: one gives shared anonymous memory huge pages when a fill is not hugepage-aligned (google/gvisor#16008), the other aligns nvproxy's mirrors of pinned application memory to huge pages (#16009). Upstream release-20261026.0 contains the first, and release-20261102.0 both. We ran both on a second card, with a rule committed before the machine existed:

gVisor	Host shared-memory THP	Ratio [95%]	Graph util
10-26	never	**0.734** [0.719, 0.748]	62.5%
11-02	never	**0.734** [0.719, 0.750]	62.1%
11-02	advise¶	**0.729** [0.717, 0.742]	61.9%

¶Exploratory, its reading committed after the first two rows and before it ran. Driver 610.57.04, vLLM 0.29.0, 12 boots per arm.

The deficit replicates on the second card, with the same idle-GPU signature (61.9–63.2% across every sandboxed deficit container on both cards). It survives the release that contains both changes. But the test does not reach the mechanism. With the host's shared-memory transparent huge pages at Ubuntu's default, `never`, gVisor cannot obtain huge pages at all; set to `advise`, the host still reported 0 kB of shared-memory huge pages mid-run. #16009's own benchmark also relies on a third change, #16215, which aligns shared anonymous mappings and is not merged upstream. So Modal's explanation is consistent with everything we measured and untested by it: its precondition, hugepage-backed GPU-shared memory, was never met.

The methodological point does not depend on that unknown. A measurement on a serverless GPU platform includes a layer the user can neither see nor pin; that layer changed during this study without notice; and the change moved the CUDA-graph path's throughput by up to a factor of two. An earlier revision attributed the effect to memory capacity, which was also wrong: both cards report **22.5 GiB**.

Three measurement choices. We report CV rather than a max/min range, because a range grows with sample size. We use fixed-prompt boots only, because grid repeats draw different prompts by design, and pooling them mixes prompt variation into boot variation (29.92% pooled, against 13.92% fixed-prompt in the same session). And a dispersion measured in one session is a draw: across sessions the same quantity ranges from 0.82% to 13.92%, a factor of seventeen, and the session's regime decides which. Within a session the boots are not bimodal — at $n = 23$ one Gaussian component fits better than two (ΔBIC -1.60) — so the two-population structure is between sessions, not within them.

The effect lives on the CUDA-graph path. vLLM emits, at every speculative boot, a warning that `CUDAGraphMode.FULL_AND_PIECEWISE` is unsupported with spec-decode for the Flash-Infer backend, and falls back to piecewise graphs. Pinning `cudagraph_mode: PIECEWISE` lands at the low end (30.04 versus a 30.57 default mean), and disabling CUDA graphs collapses the dispersion to CV 1.44% over 8 boots. `--enforce-eager` changes more than graphs — memory reservation, boot feasibility (§7) and FP8 kernel performance move with it — so on its own this

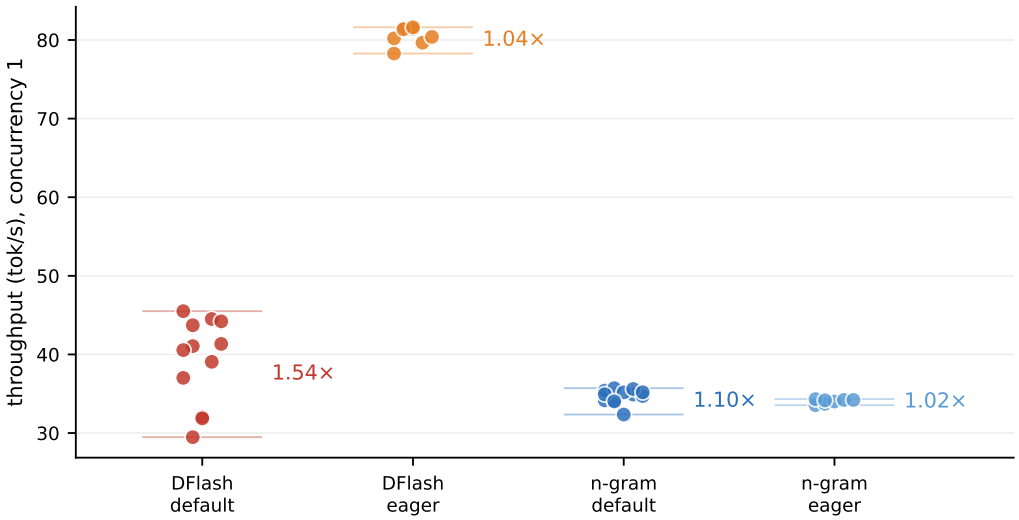


Figure 1: One dot per boot at concurrency 1, *fixed-prompt boots only* so that only the boot varies. DFlash's default arm is a wide, continuously filled distribution (CV 13.92%, $n = 12$); its eager arm and both n-gram arms are tight (1.44%, 2.67%, 0.85%). Every eager boot exceeds every default boot. *This figure plots the 2026-09-23 session only, which is a deficit session.* Nine further sessions of the identical configuration give default arms from 0.82% to 10.38% CV, seven of them at parity with their eager control, so this distribution's width is a property of the session's regime and not of the configuration; see §4. Earlier versions of this figure were wrong in three ways, recorded in the artifact (F050).

localises the effect to the graph path without naming a mechanism. The sandbox legs above say what that path pays for under gVisor.

The sharpest instance. One content-addressed cell (`dflash | bf16 | fp8_e4m3 | FLASHINFER`, graphs on) was measured in two ordinary grid runs:

Session	measurements (tok/s)	mean	τ
grid 1	28.59, 29.39, 33.81	30.60	4.183
grid 2	64.89, 71.77	68.33	4.211
ratio		**2.23**×	0.68% apart

Same `cell_id`, therefore the same configuration by construction, and a matching `argv` fingerprint. The τ difference of 0.68% is within its measured dispersion (§5). Every earlier demonstration of this effect compared *something* — different arms, different cells, boots inside a test built to show it. This compares nothing.

We report no speedup ratio from this effect. Measuring both arms inside a single run, with the default boot landing high in its range, gives 1.13 × /1.06 × /1.05× at concurrency 1/16/64. Against a low-end boot the same eager numbers give 2.19×. The ratio is a statement about where in the default arm's distribution the comparison boot happened to land, not about the flag.

What is stable, within a deficit session, is the *ordering*: CV 13.92% over 12 default boots against 1.44% over 8 eager boots, and every eager boot exceeds every default boot. **That**

ordering is regime-conditional. It holds in all 3 deficit sessions (13.92 vs 1.44; 3.73 vs 1.69; 10.38 vs 6.25) and in only 3 of the 7 parity sessions (1.53 vs 1.24; 1.95 vs 1.02; 0.82 vs 0.75; the other four invert). In the parity regime both arms sit between 0.75% and 3.12% and the comparison carries no signal. The dispersion *magnitude* is not stable and we no longer describe it as such. Turning CUDA graphs off does not reliably make speculative decoding faster by a stated factor — it makes it *reproducible*, and whatever throughput it gains is however far down the distribution the default would otherwise have fallen.

Whether it is specific to draft-model speculation is not settled. n-gram (prompt-lookup) speculation gives CV 2.67% over twelve boots against DFlash's 13.92%, and no benefit from `--enforce-eager`. That comparison carries little information. Both n-gram sessions sit in the parity regime, where DFlash itself gives 0.82–1.95%, and if n-gram entered the deficit regime at DFlash's rate of 3 in 10, two sessions would both land at parity with probability $(7/10)^2 = 0.49$. Selecting n-gram also makes vLLM fall back to its V1 model runner with async scheduling off — every n-gram log records both, and no DFlash log records either — so those sessions changed three things at once: Confound 1's shape again, in a different component. The non-speculative cell, which keeps the V2 runner and async scheduling, is the within-container control instead. As a sanity check, n-gram's acceptance is deterministic ($\tau = 1.732$ on every boot), as a prompt-matching mechanism's should be.

5 Confound 3: acceptance dispersion is small, and had not been measured

Across 12 boots of an identical configuration with identical prompts, acceptance has a coefficient of variation of **0.62%** (mean τ 4.1055, sd 0.0256, 95% CI on the mean [4.0893, 4.1218]).

Unlike the throughput dispersion, this one replicates — in both regimes. We have now measured the same quantity in every session of the sweep, across three engine releases and both throughput regimes:

Engine	Session	n	Mean τ	CV
0.29.0	2026-09-23	12	4.1055	0.62%
0.29.0	2026-09-29	12	4.1255	0.30%
0.29.0	2026-10-04	12	4.1594	0.00%
0.29.0	2026-10-31 a	12	4.1337	0.88%
0.29.0	2026-10-31 b	12	4.1564	0.26%
0.29.0	2026-10-31 c	12	4.1421	0.60%
0.29.0	2026-11-01	12	4.1675	0.11%
0.30.0	2026-09-29	12	4.1498	0.61%
0.30.0	2026-09-29 b	12	4.0702	0.00%
0.32.0	2026-10-31	12	4.1594	0.00%

Ten sessions, three engine releases: mean τ spans 4.0702–4.1675, a range of 2.4%, and no session disperses by more than 0.88%. This is the contrast that makes §4 worth reporting, and it is now the cleaner of the two findings. On the same boots, acceptance is reproducible and reproducibly quiet while throughput dispersion ranges over a factor of seventeen. Whatever varies between sessions is not changing which tokens the drafter gets accepted; it is changing how fast they are produced.

The contrast is itself regime-dependent, which sharpens it. In the three deficit sessions throughput dispersion exceeds τ dispersion by a factor of 13 to 22 where τ varies at all. In the parity sessions both quantities are below 2% and the factor falls to 1.6–7.6. So "acceptance is stable where throughput is not" is a statement about the deficit regime specifically: in the parity regime both are stable and there is no contrast to draw. The τ result holds across all ten sessions regardless; what varies is how much work it does.

A dispersion is not a detection floor. A range over a few boots (1.32% over four, **1.38%** over twelve) grows with n and encodes no variance, power or test, so it cannot mark a threshold below which effects are unfalsifiable; they are only undetected at that n. And a figure measured at $\tau \approx 4.1$ on GSM8K at 768 tokens says nothing about a comparison at $\tau \approx 2.71$ on a different prompt set at 256 tokens.

What the data supports, and what it does not. It is tempting to convert 0.62% into a detection threshold and declare the 0.7% FP8 precision effect "not detected". We decline, for a reason that is the whole point of this section.

The 12 replicated boots are DFlash, GSM8K, 768-token budget, at $\tau \approx 4.1$. The 0.7% precision effect comes from *single-boot* cells at $\tau \approx 2.71$ on a different prompt set at a 256-token budget. There are not twelve boots per precision arm; there is one. Using dispersion measured at one operating point as an uncertainty bound for a comparison made at another is precisely the transplant described above.

The defensible statement is therefore weaker and shorter: *our FP8 precision comparison is unreplicated, and cannot establish either a difference or an equivalence.* Establishing equivalence would need repeated boots per precision arm at a declared margin, which we did not run.

Acceptance is not continuously noisy. Across those 12 boots τ takes exactly **two distinct values**, 4.0914 (nine boots) and 4.1479 (three). Given fixed prompts, acceptance is near-deterministic with occasional discrete shifts, so modelling it as Gaussian measurement noise was the wrong frame independently of the sample-size error.

A withdrawn claim. An earlier version of this paper reported that backend choice perturbs τ by 2.25%. That came from one boot per backend, and it is withdrawn as unreplicated.

What we would claim, narrowly. We could not find a published dispersion estimate for τ under repeated identical boots, and effects smaller than the one we measure here have been reported as real elsewhere. The underlying practice, however, is not ours and we do not claim it. Two concurrent papers arrive at it independently, from two further directions. Kaplan [9] calibrates an LLM judge by scoring "two ordinary runs of a model given the same prompts" and adds a null condition "provably identical in distribution…whose measured difference must be zero". Pape et al. [10] reach the same prescription from cross-engine reproducibility, recommending that "researchers should avoid relying on single evaluation runs" and that results be averaged across multiple seeds. Three independent arrivals at one principle — from output quality, from backend reproducibility, and from serving throughput — is better evidence that the principle is right than any one of them alone, and we would rather say so than be found to have cited whichever we happened to read first. Our contribution is to apply it to acceptance, and to report that the dispersion is small rather than to assert a threshold.

Operating point matters. Our 0.62% is measured at $\tau \approx 4.1$, GSM8K, 768-token budget. The same configuration at a 256-token budget on a different prompt set gave $\tau \approx 2.71$ — a 55% difference from generation length and prompt distribution alone. A τ quoted without both is close to uninterpretable, and the literature quotes τ freely.

6 Confound 4: "inter-token latency" is inter-chunk latency

The first three confounds add error. This one reverses the ranking.

A streaming client measures the gap between *arrivals*. vLLM emits one SSE chunk per decoding step, and a speculative verification step emits every accepted draft token at once. So a metric computed from stream-arrival timestamps — which is how our own harness computed it, and the natural reading of a streaming API — is *inter-chunk* latency, and on a speculative server it exceeds the amortised production time per emitted token by roughly τ.

We make no claim about which specific load generators do this. We did not audit any, and saying "every load generator" without naming and versioning them would be the kind of unsupported universal this paper objects to. What we establish is the mechanism and that one harness — ours — fell into it.

Table 2: Concurrency 64, CUDA graphs on, FP8 weights, auto KV, identical prompts and SLO.

	measured gap	tokens/chunk	amortised per token	throughput
speculative (DFlash)	**93.8 ms**	**4.122**	**22.8 ms**	**2219 tok/s**
non-speculative	33.8 ms	1.007	33.6 ms	1281 tok/s

The 22.8 ms figure is **amortised production time per emitted token**, not a per-token latency: those four tokens become visible to the client together, at one instant, 93.8 ms apart. Speculation produces tokens 1.5× faster on that amortised basis and appears 2.8× slower on the recorded metric. The `tokens_per_chunk` of 4.122 tracks the measured τ of ≈4.2, as it must.

Table 3: Goodput at p95 ITL ≤ 50 ms, TTFT ≤ 1000 ms, concurrency 64.

Configuration	raw req/s	SLO met	goodput req/s
dflash | fp8 | auto	**8.31**	**0%**	**0.000**
dflash | bf16 | auto	6.13	0%	0.000
none | fp8 | auto	5.20	100%	**5.188**
none | bf16 | auto	3.37	6%	0.209

Under this objective every speculative configuration receives zero compliant goodput while producing the most tokens per second in the grid.

This is not an artifact of the 50 ms threshold. Sweeping the ITL target at TTFT ≤ 1000 ms:
At concurrency 64 the speculative configuration scores zero at every threshold we tested, while the non-speculative one reaches 99% at 50 ms. The mechanism is not a coincidence of threshold choice: the chunk cadence at that concurrency is 93.8 ms, so *any* inter-arrival target below the cadence fails by construction, and any target above it would pass trivially. At concurrency 1 and 16 the effect does not bite.

conc	mech	20 ms	35 ms	50 ms	75 ms
1	dflash	0%	64%	100%	100%
1	none	0%	89%	99%	100%
16	dflash	0%	0%	74%	99%
16	none	0%	89%	98%	100%
64	**dflash**	**0%**	**0%**	**0%**	**0%**
64	none	0%	3%	99%	100%

What this does and does not show. It shows that an SLO computed from stream-arrival gaps — which is how load generators compute ITL — inverts the ranking under speculation at the concurrency we tested. It does *not* show that all reasonable service objectives do so. We measured one workload, one arrival pattern (closed-loop fixed concurrency rather than open-loop arrival rates), and four thresholds. An objective defined over token-delivery curves, or over TTFT and end-to-end latency, would behave differently, and we did not measure chunk-size distributions.

This is not an engine defect or a bug in our harness: a server cannot deliver four tokens at four separate times when it produced them at one time. A client measuring arrival gaps will see this wherever a decoding step emits more than one token. At concurrency 1 the effect is present but does not bite — gaps are small in absolute terms and speculative cells meet the same target 100% of the time. It appears exactly where serving operates.

We do not claim the quotient is the right metric. Dividing by `tokens_per_chunk` recovers an average and discards the burstiness an ITL target exists to capture; whether a reader perceives four tokens every 94 ms as smoother than one every 34 ms is a question we did not study. Our claim is narrower: *ITL computed from stream arrivals does not measure what its name says under speculation*, and on this stack, at this concurrency, using it to gate a deployment inverts the decision. Whether a given deployment's tooling computes it that way is a question for that tooling. Prior work (§9) reports single-request latency, where the effect does not bite — consistent with it going unreported.

7 Measurements after diagnosing four confounds

We say *after diagnosing* rather than *with all four controlled*. Three are controlled: the attention backend is pinned, the execution path is carried as an axis rather than fixed, and timing is reported as chunk cadence. The fourth is not — draft-side backend selection remains uncontrolled (§3), because vLLM offers no way to pin it.

7.1 The execution-path flag is not a free control

We ran the full grid twice, 16 boots each, in independent containers, with `enforce_eager` as an *axis* rather than a pinned setting. Ratios are graphs-off over graphs-on, so below 1.0 means eager is slower.

`--enforce-eager` costs FP8-weight configurations a fifth to a half of their throughput and costs BF16 essentially nothing. Individual boots do not overlap on any FP8 row and overlap on every BF16 row, in both grids. τ is unmoved throughout.

The direction replicates on every cell; the coefficient does not, and that is itself a result. The graphs-on arm reproduces everywhere (0.95–1.06×), the eager arm reproduces for BF16

Table 4: Effect of `--enforce-eager`, two independent grids.

Configuration	conc	grid 1	grid 2
none | bf16 | auto	1	0.95×	0.96×
none | bf16 | auto	64	0.96×	0.96×
none | **fp8** | auto	1	**0.49**×	**0.68**×
none | **fp8** | auto	64	**0.66**×	**0.78**×
none | **fp8** | fp8_e4m3	1	**0.50**×	**0.66**×
dflash | bf16 | auto	1	1.13×	1.32×
dflash | **fp8** | auto	1	**0.61**×	**0.76**×

(1.00–1.05×) and does not for FP8 weights (1.22–1.40×). The eager+FP8 arm is boot-unstable — tight within a boot (20.2 and 20.5 tok/s on grid 1's two repeats) and 1.4× between them.

This study therefore found *two* unstable dimensions, each with a stable partner: on Modal, speculation is unreproducible with CUDA graphs on and stable with them off; FP8 weights are stable with graphs on and unstable with them off. Neither is visible in a single boot, and neither predicts the other.

A third constraint on the same flag: it cannot boot speculative cells with FP8 KV at all on this card, failing on an 892 MiB float32 logits buffer whose size is fixed by scheduler width times speculative width times vocabulary. Had we pinned eager — the obvious response to §4 — every FP8-weight baseline would have been reduced, every speculative FP8-KV cell would have failed to boot, and the speculation-versus-baseline speedups would have been inflated on exactly the cells this paper is about, with nothing in the output to indicate it.

7.2 A pre-registered prediction, and how it did

Before running the second grid we recorded a prediction: that the apparent ≈3× FP8-weight speculative advantage was mostly mode assignment, that it would survive but shrink to roughly 1.2–1.4×, and that FP8 rows would move least between arms.

	grid 1	grid 2
bf16 speculative, c=1, graphs off	79.5 tok/s	83.6 tok/s
FP8-weight advantage, graphs **on**	**1.69**×	**1.90**×
FP8-weight advantage, graphs **off**	**0.91**×	**1.09**×

Part 1 holds in both grids. Part 3 holds, but for the wrong reason: FP8 rows move most in absolute terms, downward, because eager penalises them. Part 2 fails: the ≈3× does shrink, but to 1.7–1.9× rather than the predicted 1.2–1.4×, and the further collapse to parity under graphs-off is a *second* effect that did not exist in our model when the prediction was written. Two effects, one of them unknown at the time, landing near a band guessed for the other. One of three held cleanly, and we report it as measured.

7.3 KV capacity

With the backend held fixed, FP8 KV cache approximately doubles token capacity, as arithmetic predicts.

Table 5: KV-cache token capacity, BF16 versus `fp8_e4m3`. Rows marked † use `auto` backend selection, where the backend differs between columns (§3) and the ratio is therefore not attributable to precision alone.

Mech	Weights	Backend	BF16 KV	FP8 KV	ratio
none	bf16	FLASHINFER	73,888	147,776	2.000×
none	bf16	TRITON_ATTN	77,168	153,872	1.994×
none	fp8	FLASHINFER	102,208	204,384	2.000×
dflash	bf16	FLASHINFER	64,784	115,920	1.789×
dflash	bf16	TRITON_ATTN	64,912	129,584	1.996×
dflash	fp8	FLASHINFER	77,616	155,232	2.000×
none	bf16	`auto`†	77,168	147,776	1.915×
dflash	bf16	`auto`†	58,080	115,920	1.996×

The FLASHINFER row is not noise, and the anomaly is speculation-specific. Every other pinned-backend row doubles to within 0.3%; `dflash | bf16 | FLASHINFER` reaches only 1.789×. Comparing the two backends that serve *both* precisions, at each precision separately:

Configuration	KV precision	between-backend spread	difference
dflash	BF16	1.002×	128 tokens
dflash	FP8	**1.118×**	**13,664 tokens**
none	BF16	1.044×	3,280 tokens
none	FP8	1.041×	6,096 tokens

For the non-speculative configuration the spread is unchanged by precision, which is what halving the bytes per token should do. For the speculative one it grows to 1.118×. So the divergence is not a property of FP8 KV in general — it appears when a draft model is also allocating KV.

We report the observation and not a mechanism: we did not profile the allocator, and block-size rounding, workspace reservation and per-model layout are all consistent with it. It is noted because it compounds §3. The backend `auto` selects under FP8 KV is FlashInfer, which is also the one realising least of the theoretical doubling on speculative configurations, so a practitioner who switches KV dtype and takes the default gets less capacity than the arithmetic promises — and gets it silently.

7.4 Task accuracy

Over 256 GSM8K problems with a 768-token budget, scored from stored generations, quantizing weights, the KV cache, or both produces no detectable change in accuracy.
Across all twelve within-mechanism paired comparisons (McNemar, exact binomial), discordant pairs number 16–23 of ≈250, p ranges from 0.238 to 1.000, and every 95% interval on the difference is contained within ±4.7 percentage points. This bounds the effect rather than demonstrating its absence: at $n = 256$ a degradation of two or three points would not be detected.

This is a task-level null, not a numerical one. Low-precision kernels in vLLM are known to disagree with each other numerically: Chen [11] finds two interchangeable INT8 GEMM implementations producing different outputs on every one of 64 sequences on SM89, with FP8 divergence growing with reduction depth where INT8 remains at parts-per-million. Nothing

Table 6: Accuracy at $n = 256$, backend pinned to FLASHINFER. No cell produced an unparseable generation.

Mech	Weights	KV cache	Accuracy	95% CI
none	bf16	auto	85.9%	81.1–89.7
none	bf16	fp8_e4m3	87.5%	82.9–91.0
none	fp8	auto	88.3%	83.8–91.7
none	fp8	fp8_e4m3	87.1%	82.4–90.7
dflash	bf16	auto	86.3%	81.6–90.0
dflash	bf16	fp8_e4m3	87.9%	83.3–91.3
dflash	fp8	auto	86.3%	81.6–90.0
dflash	fp8	fp8_e4m3	87.1%	82.4–90.7

above contradicts that, and nothing above tests it. We compare task accuracy end to end at $n = 256$; a difference in logits large enough to flip tokens can still leave a GSM8K score untouched, and our intervals are wide enough to hide several points. Read this table as "no accuracy cost we could detect at this sample size", not as evidence that FP8 arithmetic is numerically equivalent.

8 Threats to validity

- **This is a single-configuration case study.** Every result is one NVIDIA L4 (SM89), one target model (Qwen3-4B), one learned draft checkpoint, one engine release (vLLM 0.29.0), one principal workload (GSM8K), and one decoding configuration. Backend selection and CUDA-graph behaviour all depend on engine version, kernels, GPU architecture and model, so none of these findings should be assumed to transfer. What transfers is the *class* of measurement error, not the magnitudes.
- **Boot and host effects are not separated.** Runs used rented instances; we did not record GPU serial, clocks or power state, so a host-to-host component cannot be distinguished from a boot-to-boot one in §4.
- **Configuration identity is not byte identity.** Our `cell_id` hashes the declared configuration; model revisions are declared as `main`, a mutable ref, and dependencies as open ranges. We resolved the commits post hoc and their upstream modification dates precede every run, but the harness did not record them at launch. See `docs/reproducibility.md`.
- **The mechanism is unidentified.** We show that the dispersion is present with CUDA graphs enabled and collapses under eager execution. We do not show which graph, scheduler, allocator, kernel or host property is responsible, and we did not record GPU identity, so a host-to-host component is not excluded.
- **Mechanisms are not established** for the FP8-KV boot failure or the FP8-weight eager penalty. Both are observed and reproduced; neither was profiled. We state the readings we find plausible and mark them as readings.
- **Client-side saturation is argued, not measured.** A single-process benchmarking client can itself queue under load and have the delay attributed to the server [12]. Our concurrency is 64 and our generations are hundreds of milliseconds long, which places us one to two orders of magnitude below the ∼1000 QPS onset those authors report, and §6's

argument is structural rather than timing-precision dependent. We did not, however, run a client-saturation control — for instance against a zero-latency mock server — so this rests on a load-regime argument rather than on a measurement of our own harness.

- **Which claims are replicated, and which are single-session.** §4 argues that a quantity measured once is a draw, so we state where this paper meets that standard. *Replicated:* the existence of two disjoint throughput regimes and the default/eager ratio within each (eleven sessions on Modal, three engine releases, 3 deficit and 8 parity); the absence of the deficit under runc off Modal (three GCP sessions on three cards, all at parity); the sandboxed deficit (seven gVisor containers on two cards, three releases, two drivers, 0.726–0.755, with the same utilisation signature in each); acceptance dispersion (ten sessions, below 0.88% in every one); the refutation of power throttling (two deficit sessions, both arms, plus four parity sessions); the eager penalty on FP8 weights, whose direction reproduces across two independent grids while its coefficient does not, which is why we report a range. *Structural rather than statistical, and so not subject to this:* backend selection under dtype change and the ITL/chunk identity, both categorical facts read from engine logs. *Single-session, and to be read as draws:* the A10 dispersion, the 0.30.0 dispersion at each individual card, each one-factor contrast on the GCP card (runc, vLLM 0.30.0, and the backported pin: one container each), and the goodput figures at concurrency 64.
- **Grid records carry no session identity.** `probes.jsonl` records a `probe_run` per container; `sweep.jsonl` records nothing equivalent. So for any result derived from the grid — capacity, accuracy, goodput — the question §4 shows to matter, *how many sessions is this*, cannot be answered from the records. We found this by trying to ask it. The fix is one field and it is not retrospective.
- **The draft-side backend confound is acknowledged, not removed.**
- **Our retraction list covers errors we caught.** The same blindness that produced them applies to finding them. We assume undiscovered instances exist; one was found the same day we wrote the list.
- **No base rate.** Comparable studies do not report their retractions, so our count is evidence of legibility, not of unusual error-proneness.

Every finding in the artifact carries a mandatory "what this does not establish" section, and this paper inherits those limits rather than softening them.

9 Related work

None reports boot-level dispersion, and the single-request results in this table are vulnerable to the confound in §4.

On benchmarks for this exact question. SPEED-Bench [5] is the closest thing to a standard for evaluating speculative decoding, integrates with vLLM, and sweeps concurrency. It reports that synthetic inputs overestimate real-world throughput and that optimal draft length depends on batch size. It does not report dispersion across repeated identical runs, which is the axis §4 concerns, and a benchmark that does not is one more place the error term in §4 would go unnoticed.

On the accuracy axis of the same switch. Pape et al. [10] run Qwen3-4B — our target model — across five inference engines on H100 and L40, the latter sharing our SM89 architecture, with twelve seeds per configuration. They report that backend choice shifts GSM8K accuracy by up to 16.6 percentage points, and that *disabling CUDA graphs* shifts accuracy "by up to +0.15%". They measure no throughput or latency at all.

The same switch that moves their output agreement by 0.15% moves our throughput by a factor of up to two, with graph-arm dispersion of 0.82–13.92% depending on the session's regime (§4), on the same model family and GPU architecture. They are different quantities, and a careful multi-seed study probed CUDA graphs and reported a null because the quantity it measured was not the one that moves.

Chen [11] makes the complementary observation at the kernel level, also on SM89: two nominally interchangeable INT8 GEMM kernels in vLLM agree on no sequence in 64, while each reproduces itself bit-for-bit across cold restarts. Byte-identical output across boots and unreproducible throughput across boots are both properties of this stack.

On client-side measurement artifacts. Chandrasekar and Kramberger [12] show that a single-process benchmarking client becomes the bottleneck under load, adding client-side queuing delay that is then attributed to the server; they validate this against a zero-latency simulator, so the artifact is provably in the client. Their distortion sets in around 1000 QPS, where the clients they test saturate at 146–443 QPS. Our measurements run at concurrency 64 with generation latencies in the hundreds of milliseconds, one to two orders of magnitude below that onset, so we do not believe this mechanism affects our numbers — but it is the same *kind* of error as §6, in which the client's view of the token stream, rather than the server, produces the misleading metric. They do not discuss speculative decoding or multi-token chunk arrival, which is the specific case §6 concerns.

On "goodput". TurboSpec [6] optimises a quantity of that name in vLLM, so the word needs care. Theirs is the rate of *successfully generated (accepted) tokens*, used as a control signal; it involves no latency target. Ours is the rate of requests *meeting a latency service objective*. The two are different measurements sharing a name, and §6 concerns only the latter.

Two relationships deserve stating precisely rather than by omission.

Liu et al. [7] report that acceptance varies "within a request, across different requests, and across different datasets". That is variation of the *quantity*; §5 measures the *instrument* — how much a repeated identical measurement moves. The two are complementary, and conflating them would be a mistake in either direction. They also establish that increasing batch size systematically reduces SD's relative speedup, which is prior art for load-dependence and we claim no novelty there.

Kong et al. [8] validate a latency model on vLLM SD measurements to $R^2 = 0.997$ using a single-sweep methodology with no repeated runs and no discussion of CUDA graphs. Our §4 identifies an error term that such a protocol cannot see. We do not claim their results are wrong — they measure A100/H100 where we measure L4, and §4 localises the effect to draft-model speculation — only that the term is unmeasured and worth checking.

On hardware. Liu et al. use H100; Kong et al. use A100 and H100; both use FP16 KV. The regime where KV capacity actually binds, and where FP8 KV therefore has something to buy, is the smaller card. That is why we measure on L4, and it is the gap this paper occupies.

10 Artifact

Harness, raw records, derived tables and findings: https://github.com/akshathtiwari/spec-fp8-study.

Archived, with a DOI that resolves to the current version: 10.5281/zenodo.23033168. Each release is archived separately, so the version of this paper that made a claim remains retrievable after the claim is restated — which has already happened once: v1.0.0 reported the §4 dispersion as a single figure, and v1.1.0 replaced it with the two measurements and the ratio reported above.

Two scripts are worth naming. `analysis/check_provenance.py` verifies that every finding's citations resolve to stored records, that stored ids still recompute from their stored configs, and that no result rests on terminal output; it exits with the defect count. `analysis/check_paper.py` verifies that every numeric figure in this paper appears in the underlying records. Both run in `paper/build.sh` before LaTeX does, so a paper that fails its own audit is not typeset, and both have found real defects in this document — including two tables that had silently diverged from the findings they summarise.

`check_provenance.py` reports two defects, and we state them rather than suppress them: findings F009 and F014 rest on probe output predating the harness's per-measurement recording, so their numbers are not independently checkable from the repository. Neither is cited in this paper. Every other finding resolves to stored records. `paper/build.sh` stops the build on any *additional, undisclosed* defect; the two above are an explicit allowance, so the gate is against regression rather than against a nonzero count.

This is the final version. Version 2.0.0 closes the study. The repository is frozen at that tag: no further measurements will be added, and the questions left open above — the hugepage mechanism and which change between vLLM 0.29.0 and 0.30.0 removes the deficit — are stated as open rather than pursued.

What content-addressing does and does not prove. A `cell_id` guarantees two records came from the same *declared* configuration. It does not guarantee the same model bytes, engine build, or dependency versions. `argv_fingerprint` and the per-record `env` block close part of the gap; `docs/reproducibility.md` states the rest, including the resolved model commits and the five changes that would make the artifact airtight.

The entire study was run on rented per-second L4 time at a cost in the tens of dollars. The error analysis in §3–§6 is what that budget can establish, and on the evidence of §9 it is what the field is missing — rather than another single-run speedup number measured on hardware most readers cannot rent.

11 Conclusion

We set out to measure how FP8 quantization interacts with speculative decoding on an L4, and most of what we can report is what stood between the question and an answer. Each confound has the same shape: a comparison held something fixed by assumption that the system was varying underneath it. The attention backend moved with the KV dtype. Throughput moved with a sandbox the user cannot see, which on one rented card costs vLLM 0.29.0's CUDA-graph path 30% and leaves its GPU idle a third of the time. The fix Modal names for its own platform, huge pages for memory shared with the GPU, could not be exercised off it, so the last link is the platform's word rather than our measurement. Acceptance length barely moved at all, which

is why it, and not throughput, is the quantity that replicates. And the latency metric most serving benchmarks report measures stream chunks, not tokens, so under speculation it ranks configurations backwards.

Three practices would have caught the first three sooner. Measure the control in the same container as the treatment, so that a card, a host or a sandbox moves both arms together and cancels in the ratio. Replicate the session, not only the boot, because the regime is a property of the session. And record the environment a result was measured in — the backend actually selected, the host driver, the container runtime — because every correction in this study came from a variable that had been recorded and not yet read, or not recorded and had to be measured. The fourth needed only that a metric be checked against what it claims to count.

References

[1] Y. Leviathan, M. Kalman, and Y. Matias. Fast inference from transformers via speculative decoding. *ICML*, 2023. arXiv:2211.17192.

[2] C. Chen, S. Borgeaud, G. Irving, J.-B. Lespiau, L. Sifre, and J. Jumper. Accelerating large language model decoding with speculative sampling. arXiv:2302.01318, 2023.

[3] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. Gonzalez, H. Zhang, and I. Stoica. Efficient memory management for large language model serving with PagedAttention. *SOSP*, 2023.

[4] K. Cobbe et al. Training verifiers to solve math word problems. arXiv:2110.14168, 2021.

[5] T. Abramovich, M. Ashkenazi, I. Putterman, B. Chislett, T. Mitra, B. Darvish Rouhani, R. Zilberstein, and Y. Geifman. SPEED-Bench: a unified and diverse benchmark for speculative decoding. arXiv:2604.09557, 2026.

[6] X. Liu, J. Park, L. Hu, W. Kwon, Z. Li, C. Zhang, K. Du, X. Mo, K. You, A. Cheung, Z. Deng, I. Stoica, and H. Zhang. TurboSpec: closed-loop speculation control system for optimizing LLM serving goodput. arXiv:2406.14066, 2024–2025.

[7] X. Liu, J. Yu, J. Park, I. Stoica, and A. Cheung. Speculative decoding: performance or illusion? *MLSys*, 2026. arXiv:2601.11580.

[8] L. Kong, M. Flynn, M. Peng, N. Shavit, M. Kurtz, and A. Marques. An interpretable latency model for speculative decoding in LLM serving. arXiv:2605.15051, 2026.

[9] J. Kaplan. A calibrated instrument for measuring how inference optimizations affect output quality. arXiv:2609.18005, 2026.

[10] D. Pape, J. Evertz, and L. Schönherr. The silent hyperparameter: quantifying the impact of inference backends on LLM reproducibility. arXiv:2605.19537, 2026.

[11] T.-R. Chen. The integer alibi: localizing cross-kernel divergence in INT8-quantized LLM inference. arXiv:2608.13756, 2026.

[12] A. Chandrasekar and J. Kramberger. Identifying and mitigating systemic measurement bias in production LLM inference benchmarks. arXiv:2605.24217, 2026.

[13] Speculative decoding meets quantization: compatibility evaluation and hierarchical framework design. arXiv:2505.22179.

[14] SpecKV: speculative decoding under KV-cache compression. arXiv:2605.02888.

[15] QSpec: speculative decoding with complementary quantization schemes. *EMNLP*, 2025. arXiv:2410.11305.

[16] QuantSpec: self-speculative decoding with hierarchical quantized KV cache. arXiv:2502.10424.

[17] ML-SpecQD: multi-level speculative decoding with quantized drafts. arXiv:2503.13565.

[18] Quasar: quantization-aware speculative decoding. arXiv:2603.01399.

Work	What it did	Gap this addresses
Performance or Illusion? [7]	Systematic SD evaluation, vLLM on H100, CUDA graphs enabled, FP16 KV	No run-to-run variance; no FP8 KV; no SLO goodput; ITL not decomposed
Interpretable latency model [8]	Models SD latency vs request rate, vLLM on A100/H100, R^2 to 0.997	Single-sweep methodology; no repeats, error bars, or CUDA-graph discussion
Calibrated instrument [9]	Calibrates an LLM judge via a null condition; measures its per-sample noise	Same principle, applied to output quality rather than throughput or acceptance
SPEED-Bench [5]	Benchmark standardising SD evaluation, vLLM and TensorRT-LLM, throughput split across concurrencies	No run-to-run dispersion; no ITL/chunk distinction; no SLO goodput
Silent hyperparameter [10]	Backend choice shifts GSM8K accuracy by 16.6 pp across 5 engines; Qwen3-4B on H100 and L40 (SM89); 12 seeds per configuration	Measures output agreement only; no throughput or latency, so the CUDA-graph axis is probed for accuracy and not for dispersion
Integer alibi [11]	Two nominally interchangeable INT8 GEMM kernels in vLLM agree on no sequence; RTX 4090 (SM89)	Kernel-level output divergence, not serving throughput; prefill regime only
Measurement bias [12]	Client-side GIL queuing distorts latency metrics; onset at $\sim$1000 QPS	Client saturation, not engine behaviour; no speculative decoding or chunked streaming
TurboSpec [6]	Closed-loop speculation control in vLLM, optimising "goodput"	Goodput there is accepted-token rate, not SLO attainment; no latency objective
Spec. Decoding Meets Quantization [13]	EAGLE-2 with W8A8, W4A16, W4A8; single-request	No FP8, no KV-cache quantization, not batched
SpecKV [14]	Adaptive γ under compression	Algorithmic, step-level, not serving goodput
QSpec [15]	W4A4 draft + W4A16 verify, shared weights	Shared-weight special case, not FP8
QuantSpec [16]	Self-speculation, 4-bit hierarchical KV	A method, not a cross-mechanism measurement
ML-SpecQD [17], Quasar [18]	Quantized drafts, W8A8 verification	Neither reports availability or boot-level variance